

### 4.3 Design of a Combinational Circuit

Design of a combinational circuit is basically a reverse process of analyzing the circuit. Note that the term *design* means that a designer needs to come up with a logic circuit following certain criteria such as minimum cost and highest speed. In this book, logic circuits are designed based on minimum cost. That is why minimization techniques are used to reduce the number of logic gates in a circuit. This will provide lower implementation cost. In summary, a combinational circuit can be designed using three steps as follows:

- 1) **Truth Table:** If the truth table is given, go to step 2; else, determine the inputs and outputs from problem definition and then derive the truth table.

TABLE 4.1 Truth table for Figure 4.1 with input,  $Z = 1$

| Inputs |     | Outputs |      |
|--------|-----|---------|------|
| $X$    | $Y$ | $Z1$    | $Z2$ |
| 0      | 0   | 0       | 0    |
| 0      | 1   | 1       | 0    |
| 1      | 0   | 1       | 0    |
| 1      | 1   | 1       | 1    |

TABLE 4.2 Truth Table for  $F$ 

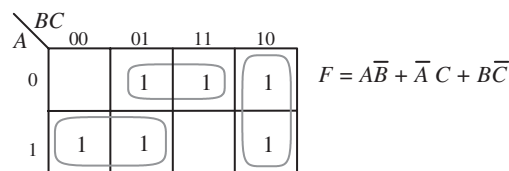
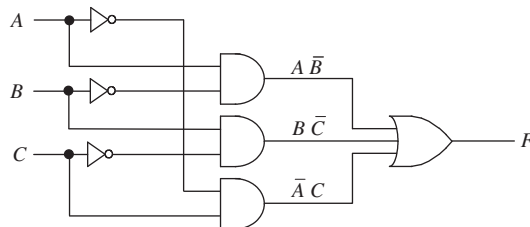
| $A$ | $B$ | $C$ | $F$ |
|-----|-----|-----|-----|
| 0   | 0   | 0   | 0   |
| 0   | 0   | 1   | 1   |
| 0   | 1   | 0   | 1   |
| 0   | 1   | 1   | 1   |
| 1   | 0   | 0   | 1   |
| 1   | 0   | 1   | 1   |
| 1   | 1   | 0   | 1   |
| 1   | 1   | 1   | 0   |

- 2) **Minimization:** Minimize the number of inputs (literals) in order to express the outputs. This reduces the number of gates and thus the implementation cost. In order to accomplish this, first use K-map(s). Note that the number of K-maps is determined by the number of outputs. If the output(s) cannot be simplified using K-map(s), then obtain output equation(s) from the truth table directly, and then simplify using Boolean identities. Simplification is also sometimes possible using the K-map(s) first and then using Boolean identities.
- 3) **Logic diagram or schematic:** Draw the logic diagram or schematic using logic gates. Note that since the terms *Logic diagram* and *Schematic* mean the same thing, they will be used interchangeably throughout this book.

In order to illustrate the design procedure, consider the following example. Suppose it is desired to design a combinational circuit with three inputs ( $A$ ,  $B$ , and  $C$ ) and one output  $F$ . The output  $F$  is one if  $A$ ,  $B$ , and  $C$  are not equal ( $A \neq B \neq C$ );  $F = 0$  otherwise. The three steps for designing the circuit is provided next:

**Step 1 Truth Table:** Since the truth table is not given, the number of inputs and outputs are determined. There are three inputs ( $A$ ,  $B$ , and  $C$ ) and one output,  $F$ . The truth table is then obtained from problem definition and is shown in Table 4.2.

**Step 2 Minimization:** Output,  $F$  in the truth table of Table 4.2 is simplified using a K-map (Figure 4.2(a)). The output equation is obtained as  $F = A\bar{B} + \bar{A}C + B\bar{C}$

(a) K-map for  $F$ (b) Logic Diagram for the output,  $F$ FIGURE 4.2 K-map and the logic diagram for  $F$ .

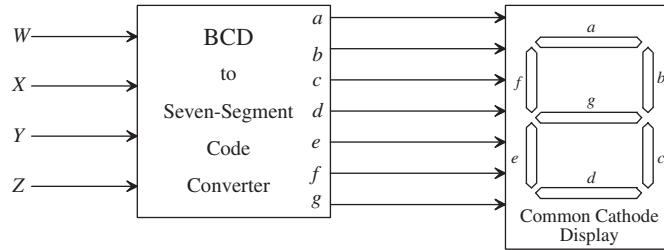


FIGURE 4.3 BCD to seven-segment code converter.

**Step 3 Logic diagram or Schematic:** Using equations of Step 2, the schematic is drawn as shown in Figure 4.2 (b).

#### 4.4 Multiple-Output Combinational Circuits

A combinational circuit may have more than one output. In such a situation, each output must be expressed as a function of the inputs. A digital circuit called the *code converter* is an example of multiple-output circuits. A code converter transforms information from one binary code to another. As an example, consider the BCD to seven-segment code converter shown in Figure 4.3. The code converter in the figure can be designed to translate a BCD input digit ( $W$ ,  $X$ ,  $Y$ , and  $Z$ ) to its corresponding seven-segment code for displaying the decimal digit. Note that seven-segment displays (both common cathode display and common anode display) are covered in Chapter 1.

The inputs  $W$ ,  $X$ ,  $Y$ , and  $Z$  can be entered into the code converter via four switches as was discussed in Chapter 1. A combinational circuit can be designed for the code converter that will translate each digit entered using four bits into appropriate seven output bits (one bit for each segment) of the display.

In this case, the code converter has four inputs and seven outputs. This code converter is commonly known as a *BCD to seven-segment decoder*. With four bits ( $W$ ,  $X$ ,  $Y$ , and  $Z$ ), there are sixteen combinations ( $0000_2$  through  $1111_2$ ) of 0's and 1's. BCD allows only 10 ( $0000_2$  through  $1001_2$ ) of these 16 combinations, so the invalid numbers ( $1010_2$  through  $1111_2$ ) will never occur for BCD and can be considered as don't cares in K-maps because it does not matter what the seven outputs ( $a$  through  $g$ ) are for these invalid combinations.

The 7447 (TTL) is a commercially available BCD to 7-segment decoder/driver chip. It is designed for driving a common-anode display. A LOW output will light a segment while a HIGH output will turn it OFF. For normal operation, the LT (Lamp test) and BI/RBO (Blanking Input / Ripple Blanking Input) must be open or connected to HIGH. The 7448 chip, on the other hand, is designed for driving a common-cathode display.

TABLE 4.3 Truth table for converting decimal digits (since common-cathode, a 1 will turn a segment ON and a 0 will turn it OFF)

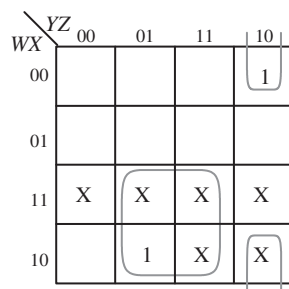
| Decimal Digit to be Displayed | BCD Input Bits |     |     |     | Seven-Segment Output Bits |     |     |     |     |     |     |
|-------------------------------|----------------|-----|-----|-----|---------------------------|-----|-----|-----|-----|-----|-----|
|                               | $W$            | $X$ | $Y$ | $Z$ | $a$                       | $b$ | $c$ | $d$ | $e$ | $f$ | $g$ |
| 2                             | 0              | 0   | 1   | 0   | 1                         | 1   | 0   | 1   | 1   | 0   | 1   |
| 4                             | 0              | 1   | 0   | 0   | 0                         | 1   | 1   | 0   | 0   | 1   | 1   |
| 9                             | 1              | 0   | 0   | 1   | 1                         | 1   | 1   | 0   | 0   | 1   | 1   |

To illustrate the design of a BCD to seven-segment decoder, consider designing a code converter for displaying the decimal digits 2, 4, and 9, and turning the display OFF for other valid BCD digits, using the diagram shown in Figure 4.3. First, it is obvious that the BCD to seven-segment decoder has four inputs and seven outputs. Three design steps are provided in the following.

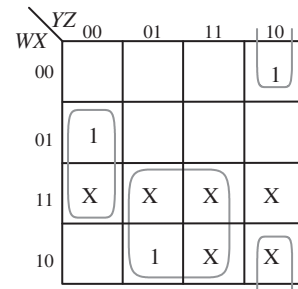
**Step 1: Truth Table** Table 4.3 shows the truth table.

**Step 2: Minimization** Figures 4.4 i) through iv) show the K-maps. Note that Seven-Segment output bits (*a* through *g*) with the same column values will need one K-map. For example, *b* and *g* with column values 111 will have the same K-map, *c* and *f* with column values 011 will have the same K-map, and finally, *d* and *e* with column values 100 will have the same K-map.

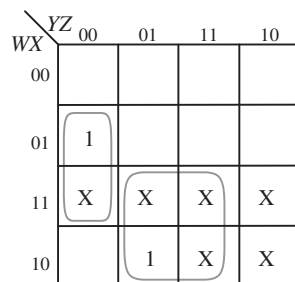
**Step 3: Schematic** Figure 4.4 v) shows the schematic. Note that the valid BCD digits that are not to be displayed (0, 1, 3, 5, 6, 7, 8), in this example, the combinational circuit for the code converter will generate 0's for the seven output bits (*a* through *g*) to turn them OFF. However, these seven bits will be don't-cares in the K-map for the invalid BCD digits 10 through 15.



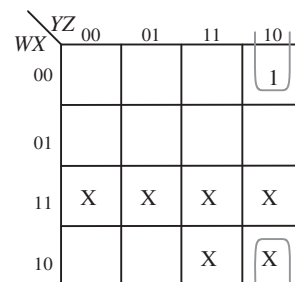
i) K-map for *a*:  $a = WZ + \bar{X}\bar{Y}\bar{Z}$



ii) K-map for *b, g*:  $b, g = X\bar{Y}\bar{Z} + WZ + \bar{X}Y\bar{Z}$



iii) K-map for *c, f*:  $c, f = X\bar{Y}\bar{Z} + WZ$

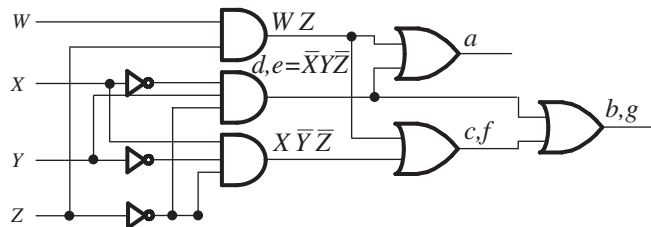


iv) K-map for *d, e*:  $d, e = \bar{X}Y\bar{Z}$

FIGURE 4.4 i) through iv) K-maps

### Example 4.1

Design a digital circuit that will convert the BCD codes for the decimal digits (0 through 9) to their Gray codes.



v) Schematic

FIGURE 4.4v) BCD to seven-segment decoder design for displaying decimal digits 2, 4, and 9.

### ***Solution***

Because both Gray code and BCD code are represented by four bits, for each decimal digit, there are four inputs and four outputs. Note that 4-bit binary combination will provide 16 ( $2^4$ ) combinations of 1's and 0's. Because only ten of these combinations (0000 through 1001) are allowed in BCD, the invalid combinations 1010 through 1111 can never occur in BCD. Therefore, these six binary inputs are considered as don't cares. This means that it does not matter what binary values are assumed by  $f_3 f_2 f_1 f_0$  for  $WXYZ = 1010$  through 1111.

#### **Step 1: Truth Table**

Table 4.4 shows the truth table.

#### **Step 2: Minimization**

Figures 4.5 a) through d) show the K-maps.

#### **Step 3: Schematic**

Figure 4.5 e) shows the schematic